Cassiopeia Gallogly

Berkeley, CA | cassiegallogly21@gmail.com
https://www.linkedin.com/in/cassie-gallogly-879199278/
https://cassiegallogly.github.io/index.html

TECHNICAL SKILLS:

Programming: C/C++, Python, MATLAB, Verilog, JavaScript

Embedded Systems: STM32, ESP32, PIC32, Arduino, UART, PCB Debugging, Sensor Integration

Robotics & Controls: Feedback Control, PID Control, Sensor Fusion, State Machines, IMU-Based Estimation

Mechanical & Software Tools: CAD (Onshape), CNC Machining, FDM 3D Printing, Soldering, Git/GitHub, LaTeX

EDUCATION

UNIVERSITY OF CALIFORNIA - BERKELEY Berkeley, CA
Master of Engineering (MEng) in Mechanical Engineering Expected May 2027
Concentration: Control of Robotic and Autonomous Systems

UNIVERSITY OF CALIFORNIA - SANTA CRUZ Santa Cruz, CA
Bachelor of Science (B.S) in Robotics Engineering August 2022 - June 2026
Minor: Electrical Engineering
GPA: 3.7/4.0, Dean's Honor List

RESEARCH EXPERIENCE

GRADUATE RESEARCHER - Liwei Lin Lab Sept 2026 - Present
- Develop and test electronics for a wearable sweat-sensing platform designed for noninvasive metabolic health monitoring.
- Troubleshoot custom PCBs through component-level voltage measurements, continuity testing, and power-distribution analysis to identify hardware faults.
- Investigate microcontroller integration, power-management circuitry, and sensor interfaces to improve prototype functionality and reliability.

UNDERGRADUATE RESEARCHER - Tactile Manipulation Lab June 2025 - Sept 2026
- Designed diamond-lattice channel geometry in CAD for soft piezoresistive tactile sensors using carbon-black/Ecoflex composites and sacrificial PVA scaffolds.
- Built a custom ESP32-based sensing circuit with multiplexed ADC measurements across 30 channel pairs and developed a Python-based real-time pressure heatmap.
- Designed experimental protocols to characterize sensor accuracy, repeatability, and dynamic force response, supporting ongoing research toward publication.

ENGINEERING PROJECTS

PROSTHETIC FOOT SUSPENSION SYSTEM Fall 2025 - June 2026
- Designed and modeled an adaptive prosthetic suspension system in CAD, iterating from initial concepts to a machined aluminum chassis.
- Performed finite element analysis (FEA) on mechanical linkage components to evaluate stress concentrations and assess structural safety under design loads.
- Selected and integrated electrical components, including sensors, servo motors, power regulation circuitry, and an ESP32 microcontroller.
- Developed embedded control functionality for sensor-based suspension adjustment across different movement conditions.

MIX-A-TRONIC | Autonomous Bartender Robot Spring 2025
- Designed and built an autonomous drink-dispensing robot integrating embedded control, motorized actuators, and sensor-based operations.

- Implemented an 11-state hierarchical state machine to coordinate cup detection, ingredient dispensing, and mechanical actuation.
- Integrated infrared cup-height sensing and automated dispensing mechanisms to support multiple drink combinations.

GESTURE-CONTROLLED WEARABLE GAME | STM32 Winter 2026

- Developed a gesture-controlled wearable glove using an STM32 microcontroller, five flex sensors, and an inertial measurement unit (IMU).
- Implemented sensor acquisition and gesture-recognition logic to translate hand movements into commands for a Python-based game.
- Integrated vibration-based haptic feedback and tested communication between the embedded hardware and game interface.

ADDITIONAL EXPERIENCE

Professional Figure Skating Coach San Jose, CA

Sharks Ice Nov 2023 – Present

- Independently plan and deliver individualized training programs for recreational and competitive figure skaters across multiple skill levels.
- Manage student schedules, client communication, and ongoing training progress while adapting instruction to individual performance goals.
- Coach athletes through competition preparation, technical development, and performance improvement

Engineering Tutor Baskin School of Engineering, UCSC

Engineering Tutor / Teaching Assistant June 2025 – Dec 2025

- Supported undergraduate engineering students in introductory robotics and technical coursework through group tutoring and individualized instruction.
- Explained programming, simulation, and engineering concepts while helping students troubleshoot technical assignments.

AWARDS

ACADEMIC: Full Graduate Scholarship - UC Berkeley, Dean's Honors List 2022 - 2026.
ATHLETIC: UC Berkeley Collegiate Figure Skating Team, US Figure Skating National Showcase Medalist 2022, US Figure Silver Skating Awards 2022, International Skating Institute Worlds 2024